Split bad primes and the tuple structure of inter-universal Teichmüller theory

A fourth report: what the source states, and what follows

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Abstract

This report collects statements about inter-universal Teichmüller theory that are either quoted from the source or proved outright, together with the computations that accompany them. It records: that a tensor packet admits place tuples of mixed reduction type, and that selecting only nonsplit bad primes does not remove them — over $\mathbb{Q}(\sqrt{5})$ a congruence family has every odd bad prime split with exactly one bad place, and for a separate explicit curve the nonsplit selection retains between 1.23% and 1.24% of the selected Tate-divisor degree; that the general-stage parameter is $\lambda = j^2 \operatorname{ord}_p(\underline{\underline{q}})$ and that when every $e_{v_i} \leq p-2$ — the condition under which Step (v) of [IUT4] leaves a place out of the set I^* it defines — [IUT4, Prop. 1.4(iii)] yields the stage margin $j^2 s - (j+2)$; where the source places the evaluation, quoted verbatim from the proof of [IUT3, Cor. 3.12]; two constraints on the indeterminacies, namely that (Ind2) acts by a unit scalar and therefore leaves the holomorphic hull unchanged, and that labels are not recoverable from a raw procession; and the identity between the two forms in which [IUT4, Thm. 1.10] prints the procession-normalized coefficient, which is $\ell(\ell+1)/12$ times the stage-one coefficient — a ratio that no statement below depends on, Proposition 10 being the statement about it. Every assertion is quoted or proved, the statement about (Ind2) is for a fixed *nonarchimedean* tuple and the one about labels for a fixed stage, and no verdict on the correctness of the theory or on *abc* is asserted.

1 Scope and status of the statements

This report accompanies the explicit initial-data construction in [First, Thm. 1], whose conditions are verified there; the expanded verification is [First-full, §§2–5]. That construction uses

$$F_0 = \mathbb{Q}(\sqrt{5}), \quad \pi = 16 + 3\sqrt{5}, \quad a = \pi^{29}, \quad E_0 : y^2 = x(x-1)(x-a), \quad \ell = 7,$$

$$F = F_0(i, E_0[15]), \qquad K = F(E_0[7]) = F_0(i, E_0[105]).$$

All multiplicative places outside $2, 7$ are selected as bad. The global cover, compatible section and cusp, and all selected local orders are verified in the cited construction, with the warrants separated there: the geometric constructions are proved and *are not verified by any program*, while the seventh-power-free clause that gives the selected orders at *all* places is certified by the exhaustive sieve of [First, §9] together with the norm identity. The field of moduli is F_0; the selected places above 211 are one good place and one bad place, with $e_{\mathrm{g}} = 1$, $e_{\mathrm{b}} = 105$ and $\operatorname{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = 29/7$.

We use **[S]** for a cited source statement or a report of an author's claim, **[P]** for a deduction with proof or a reproducible computation, and **[H]** for an unproved identification or source reading. Reporting a claim as [S] does not certify its truth. *Every assertion below is either quoted from a source or proved.* We do not claim more than that. In particular we do not claim that no reading is involved anywhere: writing $S(R)$ for "the source asserts R", $S(R)$ and R are different statements, and $S(R) \not\Rightarrow R$. Where a consequence is drawn from a quotation, what is proved is either a statement about the text or a mathematical statement whose unproved inputs are cited theorems, named as **[S]** where they are used — Propositions 9 and 3 are of the second kind. Two scopes are stated explicitly rather than left to the reader: Proposition 9 is about a *fixed* nonarchimedean tuple, and the procession argument of §6 is about a *fixed* stage. An earlier version guaranteed that nothing here was conditional on any reading; that guarantee could not stand as written (2026-09-08). The source versions are specified in the bibliography. In particular, a reference to Step (v) means the proof of Theorem 1.10 of [IUT4].

2 Place tuples and mixed reduction

Proposition 1 ([P])**.** *For a rational prime p let $\mathbb{V}_p = \{v \in \mathbb{V} : v \mid p\}$ and let A be a label set with $|A| \geq 2$. The tuple set of the local tensor packet is $\mathbb{V}_p^A$. Nonconstant place tuples exist if and only if $|\mathbb{V}_p| \geq 2$, equivalently if F_{mod} has at least two places above p. Tuples containing both chosen-good and chosen-bad places exist if and only if both kinds of places occur in $\mathbb{V}_p$.*

Proof. By [IUT1, Definition 3.1(e)], the section $\mathbb{V} \to \mathbb{V}_{\mathrm{mod}}$ restricts to a bijection above each p. Distributing the tensor product of direct sums in [IUT3, Propositions 3.1, 3.2] gives

$$\bigotimes_{\alpha \in A} \bigoplus_{v \in \mathbb{V}_p} V_{\alpha,v} = \bigoplus_{t \in \mathbb{V}_p^A} \bigotimes_{\alpha \in A} V_{\alpha,t(\alpha)}.$$

Each assertion now follows by choosing the values of a function $t : A \to \mathbb{V}_p$. No reduction hypothesis is needed for nonconstancy. □

The same dichotomy is stated in [First, Prop. 2, Cor. 3]. Thus the mechanism involving mixed place tuples is absent when $F_{\mathrm{mod}} = \mathbb{Q}$. Over a quadratic field a nonsplit rational prime has a single place, while a split prime has two; the split case can still have two good places or two bad places. Splitting alone does not supply mixed reduction. We make no claim here about the full scope of the effective results in [MFHMP]. Under Theorem 1.10 of [IUT4], chosen and actual reduction types agree outside 2ℓ, as shown next.

3 What changing the selected bad set can accomplish

Proposition 2 (the selection constraint; [P])**.** *For a fixed initial datum satisfying the additional good-reduction hypothesis of [IUT4, Theorem 1.10], the chosen bad places outside 2ℓ are exactly the actual multiplicative places there. Changing the chosen set alone cannot eliminate a mixed profile at such a prime while retaining those hypotheses.*

Proof. Definition 3.1(b) of [IUT1] permits a nonempty subset of odd multiplicative places; in particular it does not permit an actual-good place to be selected as bad. The extra hypothesis at the opening of [IUT4, Theorem 1.10, p. 22] requires actual good reduction at each chosen-good place outside 2ℓ. It therefore prohibits leaving an actual-bad place outside 2ℓ unselected. These two inclusions give the equality. The distinction between the freedom in the definition and the hypotheses for applying the theorem is essential. □

[First, Question 1, Remark 3] make the same point.

Proposition 3 (a family with no odd nonsplit bad place; [P]). *Let $a = A - \sqrt{5}$, where $A \in \mathbb{Z}$, $A \equiv 2 \pmod 5$ and $A \equiv 2 \pmod{19}$. For the Legendre curve over $F_0 = \mathbb{Q}(\sqrt{5})$, every odd rational prime below a bad place splits in F_0, with exactly one bad place above it. The set of odd nonsplit bad places is empty.*

Proof. The integral Legendre equation has $\Delta = 16a^2(a-1)^2$ and $c_4 = 16(a^2 - a + 1)$. At an odd place it has good reduction when $a \not\equiv 0, 1$, and multiplicative reduction when $a \equiv 0$ or 1, since then c_4 is a unit. For inert p, $\sqrt{5} \notin \mathbb{F}_p$, so $A - \sqrt{5}$ cannot be 0 or 1 in the residue field. At the ramified prime 5, its residue is 2. Now suppose that both places over a split odd p were bad. For a nonzero root s of $s^2 = 5$ in $\mathbb{F}_p$, the distinct values $A - s, A + s$ would have to be $0, 1$ in some order. Hence $2A = 1$ and $4s^2 = 1$ in $\mathbb{F}_p$. Thus $p \mid 19$ and $A = 10 \pmod{19}$, contrary to the hypothesis. This proves the assertion. □

The separate Hensel/CRT existence family of [First-full, Prop. 9.4] satisfies these congruences and has a bad place at 211. For its admissible members, selecting only nonsplit odd bad places would select the empty set, contrary even to the nonemptiness clause in [IUT1, Definition 3.1]. Proposition 3 is a statement about reduction for every stated A; the additional initial-data conditions for the admissible subfamily are not part of that proposition: [First-full, Prop. 9.4] supplies the arithmetic — the orders, the congruences and the seventh-power-free norm — while initial-Θ-datum status for the family is claimed in the unnumbered paragraph following it, on the strength of that draft's §4, which we do not certify here. We do not assert those conditions for arbitrary powers π^n or arbitrary A.

A different example: the numerical cost for π^{29}

The following numbers concern $a = \pi^{29} = A_{29} + B_{29}\sqrt{5}$, not the family $A - \sqrt{5}$. Exact integer arithmetic gives

$$\gcd(A_{29}, B_{29}) = 1, \qquad \gcd(A_{29} - 1, B_{29}) = 3,$$

$$R = \mathrm{N}(1-a) = 25362449986390845063221249700759698691829298140087354726503101384580,$$

and $v_2(R) = 2$, $v_3(R) = 2$, $v_5(R) = 1$, $v_7(R) = 0$. At an inert odd p, the condition $a = 0$ in the residue field requires $p \mid A_{29}, B_{29}$, which is impossible; the condition $a = 1$ requires $p \mid A_{29} - 1, B_{29}$, leaving only $p = 3$. This prime is indeed inert and bad. At 5, the residue of a is 1, giving the other nonsplit bad prime. All other odd bad primes split. Both places of a split prime cannot have $a = 0$ or both have $a = 1$, by the gcds above. If they had respectively $a = 0, 1$, the prime would divide $\mathrm{N}(a) = 211^{29}$, but the two residues of a at 211 are $0, 26$. Thus these split bad primes have mixed reduction.

Define H_{all} as the degree-normalized Tate-divisor contribution from bad places outside $2, 7$, and H_{ns} as its nonsplit part, using $1/[F_0 : \mathbb{Q}] = 1/2$. The Tate orders at odd bad places are twice the discrete orders of a or $a - 1$, so

$$H_{\mathrm{all}} = 29 \log 211 + \log(R/4), \qquad H_{\mathrm{ns}} = 2\log 3 + \log 5 = \log 45. \tag{1}$$

In the ramified case 5, the residue degree is 1 and the discrete order of $a - 1$ is 1; the factor $1/2$ cancels the Tate order 2, giving $\log 5$. The inert place at 3 has residue degree 2 and Tate order

2, giving $2\log 3$ after normalization. These are contributions to a selected divisor, not full Weil heights. The independent rational-series certificate in `scripts/check_round3.py` proves

$$309.021477380491 < H_{\mathrm{all}} < 309.021477380492,$$
$$3.806662489770 < H_{\mathrm{ns}} < 3.806662489771,$$
$$0.012318439876860 < H_{\mathrm{ns}}/H_{\mathrm{all}} < 0.012318439876861.$$

Discarding the split part would therefore discard more than 98.76% of this selected term, even if that operation were permitted. In the $A - \sqrt{5}$ family of Proposition 3, the corresponding H_{ns} is instead zero. The two statements must not be conflated.

4 The logarithm lattice and the corrected stage margin

Lemma 4 ([P])**.** *Let $p > 2$ and $K_i/\mathbb{Q}_p$ be finite extensions with $e_i \le p-2$, $0 \le i \le j$. In the tensor algebra put $R_t = \bigotimes_{i,\mathbb{Z}_p} \mathcal{O}_{K_i}$, $L_t = \bigotimes_{i,\mathbb{Z}_p} \log \mathcal{O}_{K_i}^{\times}$, and let $\widetilde{O}_t$ be the product of integer rings of its field components. For its ideal hull and degree-normalized logarithmic volume,*

$$C_t(L_t) = \Big(\bigotimes_i \varpi_i\Big)\widetilde{O}_t, \qquad \mu^{\log}(C_t(L_t)) = -\sum_i e_i^{-1}\log p,$$

where ϖ_i is a uniformizer. If $L_t \subseteq C_t(U_t)$ then $\mu^{\log}(C_t(U_t)) \ge -\sum_i e_i^{-1}\log p$.

Proof. Under the stated bound, [IUT4, Proposition 1.2(i)] gives $\log \mathcal{O}_{K_i}^{\times} = \mathfrak{m}_i = \varpi_i \mathcal{O}_{K_i}$. Thus $L_t = (\bigotimes_i \varpi_i) R_t$ and its ideal hull has the displayed form. Each embedding of the tensor algebra into a field component assigns valuation $\sum_i 1/e_i$ to the generator, with $\mathrm{ord}_p(p) = 1$. The volume convention of [IUT4, Proposition 1.4(i)] yields the equality. The inequality follows by monotonicity of the hull and its logarithmic volume. No assertion that L_t belongs to U_t before taking the hull is required. □

Odd residue characteristic and tameness alone do not suffice. For $K = \mathbb{Q}_3(\zeta_3)$, $e = 2 = p-1$ is tame. Put $\varpi = \zeta_3 - 1$, $U_r = 1 + \mathfrak{m}^r$. The quotient U_1/U_2 has order 3 and is represented by μ_3, so $U_1 = \mu_3 U_2$. The logarithm kills μ_3, and logarithm and exponential are inverse on U_2 and $\mathfrak{m}^2$ since their normalized valuations are greater than $1/(p-1)$. The remaining residue units are torsion. Consequently $\log \mathcal{O}_K^{\times} = \mathfrak{m}^2 \ne \mathfrak{m}$. This also explains why exceeding $p-2$ should not be called the wild case: that range contains tame extensions as well.

Proposition 5 (the numerical margin; [P])**.** *Under the hypotheses of Lemma 4, let the distinguished slot j be bad and put $s = \mathrm{ord}_p(\underline{q}_{t_j})$. Step (v) of [IUT4, Theorem 1.10] uses $\lambda = j^2 s$. If the target hull contains L_t, its lower bound from the lemma exceeds the expression $(-\lambda + d_I + 1)\log p$ by*

$$\big(j^2 s - (j+2)\big)\log p.$$

The difference is positive exactly when

$$j^2 s > j + 2, \qquad \textit{equivalently} \qquad \mathrm{ord}_p(q_{t_j}) > \frac{2\ell(j+2)}{j^2}. \tag{2}$$

Proof. Here every extension is tame and $d_I = (j+1) - \sum_i 1/e_i$. The choice $I^* = \varnothing$ is admissible in [IUT4, Proposition 1.4(iii)]. Subtracting its numerical expression from the lemma's bound gives $-\sum_i 1/e_i + j^2 s - d_I - 1 = j^2 s - (j+2)$. The root is $q^{1/(2\ell)}$, proving the equivalent form. □

The proposition is a statement about the expression of [IUT4, Prop. 1.4(iii)] at a given stage; whether that expression governs the actual comparison region is a separate question, not decided here. In the explicit π^{29} example, $s = 29/7$. At $j = 1$, target (g, b) gives upper expression $-226/105$, seed-hull volume $-106/105$ and margin $8/7$, after division by $\log 211$. At $j = 2$, source $(\mathrm{b}, \mathrm{b}, \mathrm{g})$ can be transported to target $(\mathrm{b}, \mathrm{g}, \mathrm{b})$ by $(1\ 2)$, fixing label 0. The corresponding values are

$$\lambda = 116/7, \qquad d_I = 208/105, \qquad a_I = 107/105,$$
$$\text{upper} = -1427/105, \quad \text{seed} = -107/105, \quad \text{margin} = 88/7.$$

This checks a conditional use of an upper expression; it does not refute Proposition 1.4 for the inputs covered by its hypotheses. The same margin, with the values $8/7$ and $88/7$ for this datum, is computed in [First, §6, Theorems 5 and 6].

Which permutations matter

For an admitted group $G \leq \mathrm{Sym}(\{0, \ldots, j\})$, a target tuple t has a preimage with good distinguished slot precisely when t_k is good for some $k \in G \cdot j$. Indeed, for $(\sigma t)_i = t_{\sigma^{-1} i}$, one has $(\sigma^{-1} t)_j = t_{\sigma j}$. Thus protecting label 0 does not block the $j = 2$ example, but fixing j does block this particular good-slot route when t_j is bad. A group may fix j and act nontrivially on other slots. Our example therefore does not require banning all nontrivial permutations as the only possible repair. Nor does this orbit calculation by itself establish admission of any permutation by the source or membership of its output in the actual union.

5 Where the evaluation lives: two columns

[S] [IUT3, Cor. 3.12, proof, (iv)] states the aim of the passage:

> "to apply the bi-coricity of the units [cf. Theorem 1.5, (iii), (iv)] in order to compute the 0-*column* Θ-pilot object in terms of the arithmetic holomorphic structure of the 1-*column*. In order to do this, one must work with units that are vertically once-shifted [i.e., lie at $(n, m-1)$] relative to the value group structures involved [i.e., which lie at (n, m)]"

and resolves the conflict between "vertical shift" and "impossibility of vertical shift" by working, on the 0-column, with structures invariant under $(0, m) \mapsto (0, m+1)$ — which is what "$(n, \circ)$" denotes. Subitem (iii) explains why a single $\Theta^{\times\mu}_{\mathrm{LGP}}$-link is used: relating units through one link and value groups through another would, the source says, render it "practically impossible to obtain conclusions that require one to relate both ... simultaneously" — that is subitem (iii). *The price of the solution adopted is recorded separately, in subitem (iv)*: it "may be implemented only at the cost of admitting the 'indeterminacy' constituted by the upper semi-compatibility of (Ind3)".

[S] Subitem (xi-b) places the output data:

> "a construction of a collection of possibilities of output data contained in $({}^{0,\circ}U^{\mathbb{Q}} \supseteq {}^{0,\circ}U \xrightarrow{\sim} {}^{1,\circ}U\ (\subseteq {}^{1,\circ}U^{\mathbb{Q}})$ — where the isomorphism '$\xrightarrow{\sim}$' arises from the permutation symmetries discussed in the final portion of Theorem 3.11, (i) — that satisfies the input prime-strip link (IPL) and simultaneous holomorphic expressibility (SHE) properties"

Subitem (xi-c) compares in the 1-column, the collection being linked by (IPL), via isomorphisms of $\mathcal{F}^{\Vdash}$-prime-strips, to the log-Kummer representation of the q-pilot object at $(1, 0)$; and (xi-d) says that the log-Kummer correspondence must be applied "in order to rectify the vertical

shift/mismatch" between the unit portion of ${}^{1,0}\mathcal{F}^{\vdash\times\mu}$ and the log-shells arising from it. *The iterates themselves are elsewhere*: [IUT3, Prop. 3.5(ii)(a)(2)] carries the m'-th iterates for $m' \geq 1$. An earlier version of this paragraph attributed "arbitrary iterates of the log-link" to (xi-d); that phrase occurs nowhere in (xi-b)–(xi-f), and joining the two places was a reading, not a quotation (internal check, 2026-09-08).

[P] Three things follow from the quoted words themselves, without settling what the passage establishes about the construction. The comparison is not confined to one column: the possibilities lie in ${}^{0,\circ}U \xrightarrow{\sim} {}^{1,\circ}U$ and the comparison is made in the 1-column. The units and the value groups sit at different vertical positions, $(n, m-1)$ and (n, m). And the isomorphism identifying the two columns is a *poly*-isomorphism furnished by the permutation symmetries of [IUT3, Thm. 3.11(i)] — a collection of isomorphisms, so no single map is selected here.

Remark 1 (five things that may not be inferred)**.** [P] A point lying in a log-shell is not thereby an output of the Θ-pilot. The relevant statement, [IUT3, Prop. 3.5(ii)(a)], is entitled *Upper Semi-compatibility* and asserts that the module in question "contains the images of the submodules of Galois invariants ... of the *groups of units*" — an inclusion in one direction, about units, not an identification with the Θ-pilot output. It also lists *two* branches through which those images arise: (1) the tensor product of the Kummer isomorphisms, and (2) their pre-composite with the m-th iterates of the log-links *for* $m \geq 1$. So the zero-iterate branch is present, and one may not treat it as absent.

[S] The source itself names the cost of the one-sided form. [IUT3, Rem. 3.10.1(i)] says that the local monoids entering $({}^{\dagger}\mathcal{F}_{\mathrm{mod}})^{\alpha}$ "are subject to a somewhat more complicated 'log-Kummer correspondence' ... that revolves around 'upper semi-compatibility', i.e., in a word, *one-sided inclusions, as opposed to precise equalities*", that "the imprecise nature of such one-sided inclusions is incompatible with the construction of" that object, and that "in particular, one cannot construct log-link-compatible isomorphisms of Frobenioids" for it — in contrast to $\mathcal{F}_{\mathrm{MOD}}$, for which the compatibility is precise.

[S] *The source states the counterpart in the same remark, and it must be read with the above.* [IUT3, Rem. 3.10.1(iii)] says that the local modules of $\mathcal{F}_{\mathrm{mod}}$ — "i.e., local fractional ideals" — are, "by contrast", "well-suited to the computation of the étale-transport indeterminacies". *The same sentence begins by saying the opposite about the other indeterminacy*: the additive nature of those modules renders F_{mod} ill-suited to the computation of Kummer-detachment indeterminacies. It continues that although "various *distortions* of these local modules arise as a result of both ... the local 'upper semi-compatibility' of Proposition 3.5, (ii), and ... the *discrepancy* between local holomorphic and mono-analytic integral structures ..., one may nevertheless compute — i.e., if one takes into account the various distortions that occur, 'estimate' — the global arithmetic degrees of objects of" $\mathcal{F}_{\mathrm{mod}}$ "by computing log-volumes ..., which are bi-coric". It then states: "*This computability is precisely the topic of Corollary 3.12 below.*" Consistently, [IUT3, Thm. 3.11(ii)] records that the tensor-packet isomorphisms of its subitem (a) — not the Frobenioid isomorphisms of the preceding paragraph — are "[precisely!] compatible" with the log-volumes, while the mutual compatibility of the Frobenioid isomorphisms holds for the subscript "MOD" "[but *not* between the isomorphisms that involve the subscript "mod"! ...]".

[P] So the one-sided inclusions above are *acknowledged by the source*: it locates them, names them as one of two sources of distortion, and asserts computability at the level of log-volumes rather than at the level of Frobenioid isomorphisms. *Whether that treatment is sufficient is a different question, and not one decided here*; what is established is that pointing at the inclusions is not by itself the exhibition of something the source has overlooked. What the quoted passages do establish is where an argument may and may not read an equality: at the level of the local modules the

relation is an inclusion, and a claim phrased as an identification of regions there is not licensed by these statements.

[P] Isomorphism class and degree do not determine a local representative. In the category of [IUT3, Ex. 3.6(ii)] an object is a collection $J = \{J_v\}_{v\in\mathbb{V}}$ of fractional ideals and, for $f \in F_{\mathrm{mod}}^{\times}$, $f \cdot J = \{f \cdot J_v\}_v$ is again an object; the product formula makes the global degree of $f \cdot J$ equal to that of J. Taking $f = 211^{-n}$ and t a place above 211 gives $\bigcup_{n\geq 0} 211^{-n} J_t = K_t$. So a fixed isomorphism class together with a fixed global degree leaves the local representative unbounded. The size of the effect is exactly a height:

Proposition 6. [P] *Let $J = \{J_v\}$ be a fractional-ideal presentation of an arithmetic line bundle over F_0 and $b \in F_0^{\times}$, so that J and bJ present the same bundle. Write E for the normalized global log-volume and $C(-)$ for the place-wise hull. Then $E(bJ) = E(J)$, while*

$$E\big(C(J \cup bJ)\big) - E(J) = \frac{1}{[F_0 : \mathbb{Q}]} \sum_v n_v \log \max(1, |b|_v) = h(b),$$

the absolute logarithmic Weil height of b. For $b = 211^{-n}$ over $F_0 = \mathbb{Q}(\sqrt{5})$ this is $n \log 211$.

Proof. At a finite place the two fractional ideals are comparable by inclusion and the hull is the larger; at an infinite place they are balls with the same centre and the hull is the larger. So the local increment over J_v is $\log \max(1, |b|_v)$, and the weighted sum is $h(b)$. The difference of the individual valuations is $\sum_v n_v \log |b|_v / [F_0 : \mathbb{Q}] = 0$ by the product formula. For $b = 211^{-n}$ the two places above 211 each contribute $n \log 211$ with weight $\frac{1}{2}$ and the two real places contribute 0. □

[S] *The source fixes this by a definition, and says what happens without it.* [IUT3, Rem. 3.9.5(i)] defines the holomorphic hull *in two branches*, for a region already assumed to contain a relatively compact subset of log-volume $> -\infty$: when the region "is relatively compact" the hull is the smallest set of the form $\lambda \cdot \mathcal{O}$ containing it, with the components of λ nonzero; and otherwise — "*If ${}^{\alpha}U \ldots$ is not relatively compact, then we define the holomorphic hull of ${}^{\alpha}U \ldots$ to be $I^{\mathbb{Q}}({}^{\alpha}\mathcal{F}_{v_{\mathbb{Q}}})$*" — it is the whole ambient object. *The standing hypothesis is not decoration*: without it there need be no smallest containing $\lambda \cdot \mathcal{O}$ at all, since $U = \{0\} \subset \mathbb{Q}_p$ is compact and lies in every $p^n\mathbb{Z}_p$. [P] So relative compactness is not a convenience: in the second branch the hull is still defined, but it gives no finite upper bound on the log-volume. An earlier version said the hull was defined "only when" the region is relatively compact and that it then "carries no information at all"; both overstated the source.

[P] *The scalar value of a hull does not determine the value of the transported hull.* Work inside the good completion K_{g} of the datum of [First-full]. *That completion is unramified of degree* 24 *over* $\mathbb{Q}_{211}$, which we verify here rather than cite: 211 splits in F_0, and at the good place $\sqrt{5} \equiv 146$, so $a = \pi^{29} \equiv 26$; the reduced curve $y^2 = x(x-1)(x-26)$ over $\mathbb{F}_{211}$ has $a_{211} = 24$ and 188 points. Since 105 is invertible there, Néron–Ogg–Shafarevich makes $K_{\mathrm{g}}/\mathbb{Q}_{211}$ unramified, of degree the order of Frobenius. That order is the order of the companion matrix because $X^2 - 24X + 211$ is separable at each of $3, 5, 7$ — its discriminant -268 is $2, 2, 5$ there — so Frobenius is conjugate to it rather than scalar or unipotent. The companion matrix of $X^2 - 24X + 211$ has order 4 on $E[3]$, 3 on $E[5]$ and 8 on $E[7]$, hence 24 on $E[105]$, and $211 \equiv 3 \pmod 4$ makes adjoining i cost 2, with $\mathrm{lcm}(24, 2) = 24$. *The coincidence with $a_{211} = 24$ is accidental.* So $e_{\mathrm{g}} = 1 < p - 1$ and $\exp, \log$ are mutually inverse on $1 + p\mathcal{O}$. Normalize $\mu(\mathcal{O}) = 1$ and write $\nu(U) = \frac{1}{24} \log \mu(U)$, $E(U) = \nu(C(U))$ with C the hull.

Proposition 7. *Put $U_r = 1 + p^r\mathcal{O}$ for $r \geq 1$. Then $E(U_r) = 0$ for every r, while $E(\log U_r) = -r \log p$. Consequently there is no function $g : \mathbb{R} \to \mathbb{R}$ with $E(\log U) = g(E(U))$ for all these U —*

with or without any linearity assumption. Adding the raw volume does not repair this: $U = 1+p^3\mathcal{O}$ *and* $V = \exp(p)\,U$ *satisfy* $\nu(U) = \nu(V) = -3\log p$ *and* $E(U) = E(V) = 0$*, yet* $E(\log U) = -3\log p$ *and* $E(\log V) = -\log p$*.*

Proof. U_r contains 1 and lies in $\mathcal{O}$, so its hull is $\mathcal{O}$ and $E(U_r) = 0$. Since $r \geq 1 > e_{\mathrm{g}}/(p-1)$, log carries U_r isometrically onto $p^r\mathcal{O}$, which is already of the form $\lambda\mathcal{O}$; as $\mu(p^r\mathcal{O}) = p^{-24r}$, $E(\log U_r) = -r\log p$. The inputs are all 0 and the outputs are pairwise distinct, so no g exists; and the outputs are unbounded below, so no two-sided error term depending on the input alone repairs it either. For the second pair, $\exp(p)$ is a unit, so $V \subseteq \mathcal{O}$ contains a unit and $C(V) = \mathcal{O}$, while multiplication by a unit preserves μ. But $\log V = p + p^3\mathcal{O}$ contains elements of valuation exactly 1 and lies in $p\mathcal{O}$, so $C(\log V) = p\mathcal{O}$. □

[S] *The source states the restriction that this respects.* [IUT3, Rem. 3.9.6] says that the log-link compatibility "amounts to a coincidence of log-volumes — *not of arbitrary regions* that appear in the domain and codomain of the log-link, but rather — of certain types of '*sufficiently small*' regions". [P] Proposition 7 is therefore not in tension with [IUT3]: it exhibits what is lost if the compatibility is read as a statement about hull scalars. *The point is not only the smallness restriction*: the U_r above are small, and E is still constant on them, because it is the passage to the hull — not the size of the region — that discards what the raw log-volume ν records.

[S] *The pre-log-shell is the image of the invariants, not the invariants of the quotient.* [IUT3, Def. 1.1(i)] defines it as "the image in $\Psi^{\sim}_{\dagger\mathcal{F}_v}$ of the submonoid of $G_v(^{\dagger}\Pi_v)$-invariants of $\Psi^{\times}_{\dagger\mathcal{F}_v}$", and records that this "constitutes a *compact* topological module"; the log-shell is that, times $(p_v^*)^{-1}$.

Proposition 8. [P] *Let* K_v *be a finite extension of* $\mathbb{Q}_p$*,* $\bar{K}_v$ *an algebraic closure,* $G = G_{K_v}$*,* $\bar{U} = \mathcal{O}^{\times}_{\bar{K}_v}$*, and* $\mu_\infty \subset \bar{U}$ *the roots of unity. Then*

$$\log \operatorname{im}\big(\bar{U}^G \to \bar{U}/\mu_\infty\big) = \log \mathcal{O}^{\times}_{K_v}, \qquad \log\big((\bar{U}/\mu_\infty)^G\big) = K_v.$$

The first is a full lattice in K_v*, hence compact; the second is not.*

Proof. On units of an algebraic closure the kernel of log is exactly μ_∞: an element lies in some finite extension, and after removing its finite-order residue and passing to a high enough p-power it lies in a neighbourhood on which exp and log are mutually inverse, so $\log z = 0$ forces that power to be 1. It is surjective onto $\bar{K}_v$: given x, choose r with $p^r x$ in the domain of convergence of exp inside a finite extension containing x, put $w = \exp(p^r x)$ and let z be a p^r-th root of w in $\bar{K}_v$; then z is a unit and $\log z = p^{-r}\log w = x$. So $\log : \bar{U}/\mu_\infty \to \bar{K}_v$ is a G-equivariant isomorphism, and taking invariants gives K_v on the right. The image of the invariant units is $\log \mathcal{O}^{\times}_{K_v}$ by definition. □

[P] So reading the pre-log-shell as the invariant part of the quotient rather than as the image of the invariants replaces a compact lattice by the whole local field, and any log-volume bound stated in terms of it becomes vacuous. This is not a defect of [IUT3], which takes the image and says so; it measures what the other reading would cost.

6 Two constraints on the indeterminacies

[P] (Ind2) *acts by a unit scalar* — Proposition 9 below, at a fixed nonarchimedean tuple. *The classification of the isometries is not ours*: [MT, Thm. F, and Thm. 7.4] state that the natural $\mathbb{Z}_p^{\times} \to \mathrm{Ism}(G_K)$ is bijective under the hypotheses recalled next, and the authors summarize this as "(Ind2) is relatively small". What is left to this report is to check those hypotheses for a finite

extension of $\mathbb{Q}_p$ and to draw the consequence for the hull. The identification is in the sources too: [IUT3, Thm. 3.11(i)] describes (Ind2) as "the indeterminacies induced by the action of independent copies of Ism ... on each of the direct summands of the $j+1$ factors", and the introduction of [MT] states, *before* Theorem F, that "(Ind2) consists of these isometries". [S] [MT, Def. 7.3] defines $\mathrm{Ism}(G_K)$ as the group of G_K-equivariant automorphisms of $U^{\mu}_{K^{\mathrm{sep}}}$ that preserve the lattice U^{μ}_{L} for *each* finite separable $K \subseteq L \subseteq K^{\mathrm{sep}}$ — not merely one lattice — and notes a natural injection $\mathbb{Z}_p^{\times} \hookrightarrow \mathrm{Ism}(G_K)$. Their Theorem 7.4 states that *if G_k is not a pro-prime-to-p group*, that injection is bijective; their Remark 7.3.1 states that *if k is finite*, this $\mathrm{Ism}(G_K)$ is compatible with the one of [IUT2, Ex. 1.8(iv)]. The authors themselves write that "Theorem F implies that (Ind2) is relatively small".

[P] Both hypotheses hold in the case we use. For K a finite extension of $\mathbb{Q}_p$ the residue field k is finite, so Remark 7.3.1 applies; and $G_k \cong \widehat{\mathbb{Z}}$ has $\mathbb{Z}_p$ as a quotient, so G_k is not pro-prime-to-p and Theorem 7.4 applies. Hence in this case $\mathrm{Ism}(G_K) = \mathbb{Z}_p^{\times}$.

Proposition 9. [P] *At a fixed* nonarchimedean *tuple t, let C_t denote the holomorphic hull of [IUT3, Rem. 3.9.5(i)] in* both *of its branches: the smallest $\lambda \cdot \mathcal{O}$ containing the region when that region is relatively compact, and the whole ambient object when it is not. Assume that U contains a relatively compact subset whose log-volume is greater than $-\infty$, as required by that definition. Then for every g in the group above, $C_t(g_t U) = C_t(U)$ and $C_t\big(\bigcup_g g_t U\big) = C_t(U)$.*

Proof. By the previous paragraph the action at t is $g_t = u_t\,\mathrm{id}$ with $u_t \in \mathbb{Z}_p^{\times}$, so each g_t is an isometry: it preserves valuations componentwise, hence preserves relative compactness and the family of sets $\lambda \cdot \mathcal{O}$.

Relatively compact branch. If U is relatively compact then so is $g_t U$, and $U \subseteq \lambda \cdot \mathcal{O}$ if and only if $g_t U \subseteq \lambda \cdot \mathcal{O}$; the two therefore have the same smallest such $\lambda \cdot \mathcal{O}$, so $C_t(g_t U) = C_t(U)$. The union $\bigcup_g g_t U$ lies in that same $\lambda \cdot \mathcal{O}$ and contains U, so it is relatively compact with the same hull.

Non-relatively-compact branch. If U is not relatively compact then neither is $g_t U$ nor the union, and all three hulls are the whole ambient object by definition.

An earlier version identified the hull with the B_t-module generated by the region. That identification does not hold in general — the module generated by a region with full support is not of the form $\lambda \cdot \mathcal{O}$ — and it is not needed: the argument above uses only that g_t is an isometry. □

In the relatively compact branch, this is proved using the same family of $\lambda\mathcal{O}$ containers; in the other branch, it follows from the definition of the hull. It concerns (Ind2) alone: it does not extend to (Ind1), (Ind3), the input fibre or the q-comparison, and it does not say that an arbitrary automorphism of G_K acts as a ring isomorphism. [S] That (Ind1) and (Ind2) preserve log-volumes was stated earlier by Takkun-kohinata/IUT_LEAN [TK] (15 July 2026, item (F2)) and by Morane [Mor, §8.2] (6 August 2026). Proposition 9 concerns the hull, and takes the unit-scalar form of (Ind2) from [MT].

[P] *Labels are not recoverable from a raw procession.* In a procession whose entries are copies of one prime-strip, the internal symmetric group of any stage acts while fixing every other stage, including the singleton stage: by [IUT1, §0] a capsule-full poly-morphism is a fixed injection together with *all* collections of isomorphisms, and by [IUT1, Def. 4.10] each arrow of a procession is the collection of all such, so the automorphisms act stage-wise independently, with the full symmetric group at each stage. The only subsets *naturally selectable* — that is, invariant under every automorphism of the raw procession — are therefore $\varnothing$ and the whole. [S] The source states the conclusion directly: [IUT1, Prop. 6.9(i)] records that "there are precisely n possibilities" for the element of $|\mathbb{F}_\ell|$ to which a given index of the n-capsule corresponds. Hence one may not deduce,

from the fixing of the singleton $\mathcal{D}_0$, that the label 0 is fixed inside the larger capsules; and a coefficient computation for a subgroup does not by itself exhibit a repair, which would require additional structure surviving in the actual weakened input together with a proof that the choice is natural. This concerns copies of one strip, not processions with mutually non-isomorphic entries.

7 The procession-normalized coefficient

[S] In Step (v) of the proof of [IUT4, Thm. 1.10] the weighted average upper bound carries the term $-\frac{j^2}{2\ell}\log(\mathfrak{q}_{v_\mathbb{Q}})$ — the coefficient is *negative*, and what is printed below are its magnitudes — at stage j — on the *degree of the q-parameter divisor*, not on its 2ℓ-th root: [IUT4, p. 23] records that $|\log(\underline{\underline{q}})|$ "is equal to" $\frac{1}{2\ell}\log(\mathfrak{q})$, so the coefficient on $\log(\underline{\underline{q}})$ at stage j is j^2. An earlier version wrote $\log(\underline{\underline{q}})$ in both places, conflating the two by a factor of 2ℓ. The source then says: "it remains to compute the average over $j \in \mathbb{F}_\ell^*$. By averaging over $j \in \{1, \dots, \ell^* = \frac{\ell-1}{2}\}$ and applying (E1), (E2), we obtain the 'procession-normalized upper bound'" — in which the magnitude of the coefficient on $\log(\mathfrak{q}_{v_\mathbb{Q}})$ is printed first as $\dfrac{(2\ell^*+1)(\ell^*+1)}{12\ell}$ and then as $\dfrac{\ell+1}{24}$.

Proposition 10. [P] *The two printed forms agree, and the average of the stage coefficients equals* $\dfrac{\ell(\ell+1)}{12}$ *times the stage-one coefficient:*

$$\frac{1}{\ell^*}\sum_{j=1}^{\ell^*}\frac{j^2}{2\ell} = \frac{(2\ell^*+1)(\ell^*+1)}{12\ell} = \frac{\ell+1}{24} = \frac{\ell(\ell+1)}{12}\cdot\frac{1}{2\ell}.$$

Equivalently, the stage-one coefficient is $\chi := \dfrac{12}{\ell(\ell+1)}$ *times the averaged one; at* $\ell = 7$, $\chi = \frac{3}{14}$ *and the averaged coefficient is* $\frac{1}{3}$.

Proof. $\sum_{j=1}^{m} j^2 = m(m+1)(2m+1)/6$ with $m = \ell^*$ gives the first equality after dividing by $\ell^* \cdot 2\ell$. Substituting $\ell^* = (\ell-1)/2$ gives $2\ell^*+1 = \ell$ and $\ell^*+1 = (\ell+1)/2$, whence $(2\ell^*+1)(\ell^*+1)/(12\ell) = (\ell+1)/24$. The last equality is $(\ell+1)/24 = \big(\ell(\ell+1)/12\big)\cdot\big(1/(2\ell)\big)$. □

This is an identity between two quantities printed in the source; it is not a statement about which region either coefficient is evaluated on. [S] The same ratio, between the averaged degree of the Θ-pilot divisor and the degree of the q-pilot divisor, is in [DHPS, §3.8, (3.3)] (2020), and [DHPS, Rem. 3.8.3(1)] relates it to the argument of Scholze and Stix.

[S] In [SS, §1.3, p. 4], formula (1.5) is introduced as holding "in essential approximation". Its coefficient on $\deg(q_E)$ is $\frac{(\ell^*+1)(2\ell^*+1)}{12\ell}$, the form printed in [IUT4, Thm. 1.10, Step (v)]. The *unnumbered* display immediately after (1.5), obtained there by substitution into (1.4), is

$$\tfrac{1}{6}\deg(q_E) \;\lessapprox\; \Big(1 - \tfrac{12}{\ell(\ell+1)}\Big)^{-1}\cdot\mathfrak{d}(P).$$

The following sentence in [SS, p. 4] retains an error, handled by adding a term linear in ℓ on the right. The display is not an error-free inequality. [P] *That factor is* χ: the ratio of the stage-one coefficient $1/(2\ell)$ to the averaged $(\ell+1)/24$ is $12/(\ell(\ell+1))$. Proposition 10 thus identifies the averaged Tate-divisor coefficient under these conventions. It does not identify the full estimates, their positive or error terms, or the regions to which they apply.

[S] *The source's own corresponding constant is a different one.* In assembling the local bounds of Steps (v)–(vii), [IUT4, Thm. 1.10, proof Step (viii), p. 30] uses $\frac{\ell+1}{4}\cdot\frac{1}{6}\cdot\frac{12}{\ell^2} \geq \frac{1}{2\ell}$, that is $12/\ell^2$,

not $12/(\ell(\ell+1))$. [P] The two are not equal, and $12/\ell^2 > 12/(\ell(\ell+1))$; the inequality just quoted has slack $\frac{\ell+1}{2\ell^2} - \frac{1}{2\ell} = \frac{1}{2\ell^2}$. [S] The same rearrangement, with the same slack $\frac{1}{2\ell^2}$, is written out in [JoshiIV, p. 70]. This is an exact comparison of these two scalar constants; the display in [SS] still has the approximation and error just stated. *What this records is that the averaging coefficient is common to both and the final constant is not. It decides nothing about [IUT3, Cor. 3.12] or about the objection of* [SS].

Neither coefficient is used in any other statement of this report.

Remark 2 (on explicit formulations)**.** [S] [DH, §4.10, eq. (4.4)] writes the multiradial representation of the Θ-pilot region as $U_\Theta := (\mathrm{Ind2})\big((\mathrm{Ind1})\big((\mathcal{O}_L(-P_\Theta))^{(\mathrm{Ind3})}\big)\big) \subset L$, and [DH, Rem. 4.10.1] states that "the order of operations and formulas for the indeterminacies do not appear in" Tan's note or in [IUT3], [IUT4]. The (Ind3) ingredient is moreover an assumed bound: [DH, §4.9] writes "we assume a bound on $\mathcal{O}_L(-P_\Theta)^{(\mathrm{Ind3})}$", and the introduction to [DH, §4] describes it as "a formula for an upper bound on the (Ind3) indeterminacies".

[S] [DH, p. 3] states, on the other hand, that the ordering in their own (1.2) is deliberate, and reports that Mochizuki considers (Ind1) and (Ind2) together. [P] One consequence, and one non-consequence. Exhibiting a map inside an over-large group is not a lower bound for the actual indeterminacy, since the bound is an upper one. But a counterexample to a formulation of this shape is not *thereby* a counterexample to [IUT3] either: that would require the equivalence of the two formulations, which is not established here and which the two passages above do not settle in either direction. An earlier version inferred from the first passage that the source is *not committed* to the ordering; that does not follow from a statement about where formulas appear .

Corrections and reproducibility

[P], added in version 1.0.4. [First] has withdrawn, in its version 0.1.7, the source application under which the values 8/7 and 88/7 were read as an incompatibility with [IUT4]; see [First, §7] for the source locators and for a proposition limiting the scope of that withdrawal to the mechanism used there. *No statement of the present report used that application*: the passage computing those values states that whether the expression of [IUT4, Prop. 1.4(iii)] governs the actual comparison region is a separate question, not decided here, and that the computation does not refute that proposition for the inputs covered by its hypotheses. The numerical values are unchanged, and nothing in this report is withdrawn.

[P], added in version 1.0.5. This version cites earlier work: the ratio of Proposition 10 in [DHPS], the slack of the rearrangement in [JoshiIV], and the volume preservation by (Ind1), (Ind2) in [TK] and [Mor]. It also points to the corresponding statements of [First] at Propositions 1, 2 and 5, and updates the version cited for [First-full]. No statement or number is changed.

Earlier drafts omitted j^2 at stage 2, producing the incorrect claimed margin 1/7; the corrected margin is 88/7. They also confused nonconstant place tuples with mixed reduction, inferred independent probabilities 2^{-r} from a splitting density, asserted the logarithm-lattice identity for every tame odd place, and claimed that every nontrivial permutation would have to be forbidden. Those inferences are withdrawn; the proofs and restrictions above replace them. The earlier comparison of 1/3 with 2/3 also used an incorrect weighting. The normalized bound coefficients are not used in any statement above except Proposition 10, which is the identity between them.

Corrections in the accompanying initial-data account include $46080 = 2^{10}3^2 5$, the exceptional invariant $2^{14}31^3/5^3$, selection of all multiplicative places outside $2, 7$ including the place over 59, and the split-multiplicative argument after adjoining i. Full rational 2-torsion does not itself exclude a

nonsplit twist. The mistaken suggestion that all four exceptional invariants are algebraic integers is also retained as an error record in the first report.

Every passage quoted above was checked against the English original of the version listed in the bibliography. *The two steps are different*: the shipped `scripts/note3_quotes.py` *locates* a candidate passage by a word-sequence match that drops mathematics and ignores punctuation, and reports twelve of the forty-one entries as too short to locate at all; verbatim identity was then checked by reading. The mechanical step used string match on the extracted text and, where text extraction interleaves floats — which it does on p. 150 of [IUT3] — by reading the rendered page. Two conventions of the extraction are worth recording, since each produced a false mismatch before being accounted for: a page's text may contain control characters inside a word, and the source uses double quotation marks where a quotation reproduced here uses single ones.

The exact scripts released with the present report — revised copies of those published with [First], certifying the same numbers and additionally refusing to run with assertions disabled — verify the displayed integer and rational arithmetic, including rigorous logarithm intervals and the normalization identities. They do not verify source interpretations. The finite Lean model checks a distinction between invariance, orbit membership and hulls of unions; it is not a model of the entire source theory. Large language models were used in drafting, computations and source review. No independent human verification or absence of further errors is asserted.

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